CAD for VLSI Design
Programming Assignment 1

Bonus Project Report

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1. **The answer of each question**

   a. **Q1: Total cell area**

      i. 16nm area: 20.736

```
Number of ports:                          43
Number of nets:                          117
Number of cells:                          81
Number of combinational cells:            81
Number of sequential cells:                0
Number of macros/black boxes:              0
Number of buf/inv:                        16
Number of references:                     19

Combinational area:               20.736001
Buf/Inv area:                      2.488320
Noncombinational area:             0.000000
Macro/Black Box area:              0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  20.736001
Total area:                     undefined
1
```

      ii. 130 nm area: 626.34

```
Number of ports:                          43
Number of nets:                          111
Number of cells:                          75
Number of combinational cells:            75
Number of sequential cells:                0
Number of macros/black boxes:              0
Number of buf/inv:                        13
Number of references:                     23

Combinational area:               626.340607
Buf/Inv area:                      44.132399
Noncombinational area:             0.000000
Macro/Black Box area:              0.000000
Net Interconnect area:      undefined  (No wire load specified)

Total cell area:                  626.340607
Total area:                     undefined
1
```

   b. **Q2: Compare two values. Which one is smaller?**

      依照 area report 可以看出，.16nm 的合成結果的面積遠小於 130nm 的合成結果，這是很合理的結果，畢竟在 W/L 以及其他條件都相同的情況下，電晶體的 L 越小，W 就越小，對於 layout 的總面積也就越小，故可證明合成結果正確且合理。

   c. **Q3: Total power**

      i. 16nm power: total power=8.65*10^-3 mW

```
Global Operating Voltage = 0.72
Power-specific unit information :
   Voltage Units = 1V
   Capacitance Units = 1.000000pf
   Time Units = 1ns
   Dynamic Power Units = 1mW    (derived from V,C,T units)
   Leakage Power Units = 1nW


  Cell Internal Power  =   4.1862 uW   (48%)
  Net Switching Power  =   4.4634 uW   (52%)
                   ---------
Total Dynamic Power    =   8.6496 uW  (100%)

Cell Leakage Power     = 331.3139 pW

Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

                 Internal       Switching        Leakage          Total
Power Group      Power          Power            Power            Power   (  %   ) Attrs
--------------------------------------------------------------------------------------------------
io_pad           0.0000         0.0000           0.0000           0.0000  (  0.00%)
memory           0.0000         0.0000           0.0000           0.0000  (  0.00%)
black_box        0.0000         0.0000           0.0000           0.0000  (  0.00%)
clock_network    0.0000         0.0000           0.0000           0.0000  (  0.00%)
register         0.0000         0.0000           0.0000           0.0000  (  0.00%)
sequential       0.0000         0.0000           0.0000           0.0000  (  0.00%)
combinational    4.1862e-03     4.4634e-03       0.3313           8.6499e-03  ( 100.00%)
--------------------------------------------------------------------------------------------------
Total            4.1862e-03 mW  4.4634e-03 mW    0.3313 nW        8.6499e-03 mW
1
```

ii. 130nm power=9.55*10^-2 mW

```
Global Operating Voltage = 1.08
Power-specific unit information :
   Voltage Units = 1V
   Capacitance Units = 1.000000pf
   Time Units = 1ns
   Dynamic Power Units = 1mW    (derived from V,C,T units)
   Leakage Power Units = 1pW


  Cell Internal Power  =  65.5419 uW   (69%)
  Net Switching Power  =  29.9612 uW   (31%)
                   ---------
Total Dynamic Power    =  95.5030 uW  (100%)

Cell Leakage Power     =  17.4919 nW

Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

                 Internal       Switching        Leakage          Total
Power Group      Power          Power            Power            Power   (  %   ) Attrs
--------------------------------------------------------------------------------------------------
io_pad           0.0000         0.0000           0.0000           0.0000  (  0.00%)
memory           0.0000         0.0000           0.0000           0.0000  (  0.00%)
black_box        0.0000         0.0000           0.0000           0.0000  (  0.00%)
clock_network    0.0000         0.0000           0.0000           0.0000  (  0.00%)
register         0.0000         0.0000           0.0000           0.0000  (  0.00%)
sequential       0.0000         0.0000           0.0000           0.0000  (  0.00%)
combinational    6.5542e-02     2.9961e-02       1.7492e+04       9.5521e-02  ( 100.00%)
--------------------------------------------------------------------------------------------------
Total            6.5542e-02 mW  2.9961e-02 mW    1.7492e+04 pW    9.5521e-02 mW
1
```

d. Q4: Compare two values. Which one is smaller?

由合成結果可知，130nm 的 gate-level netlist 的 power 估計結果大於 16nm 的 gate-level netlist 的 power，這是合理的結果，因為 Power=P_{static}+$P_{switching}$+$P_{short\ circuit}$，而 $P_{switching}$ = α*C_L*f*Vdd^2，$P_{short\ circuit}$=β*$(Vdd-2Vt)^3$*f*tr/12，故在 W/L 固定的情況下，$P_{switching}$ 正比於 load capacitance，又 load capacitance 正比於電晶體的 W，故 130nm 的 power 會比 16nm 的 power 大，合理且正確。

$$P_{sc} = \frac{\beta}{12}(V_{DD} - 2V_t)^3 \tau f \, (t_r = t_f = \tau)$$

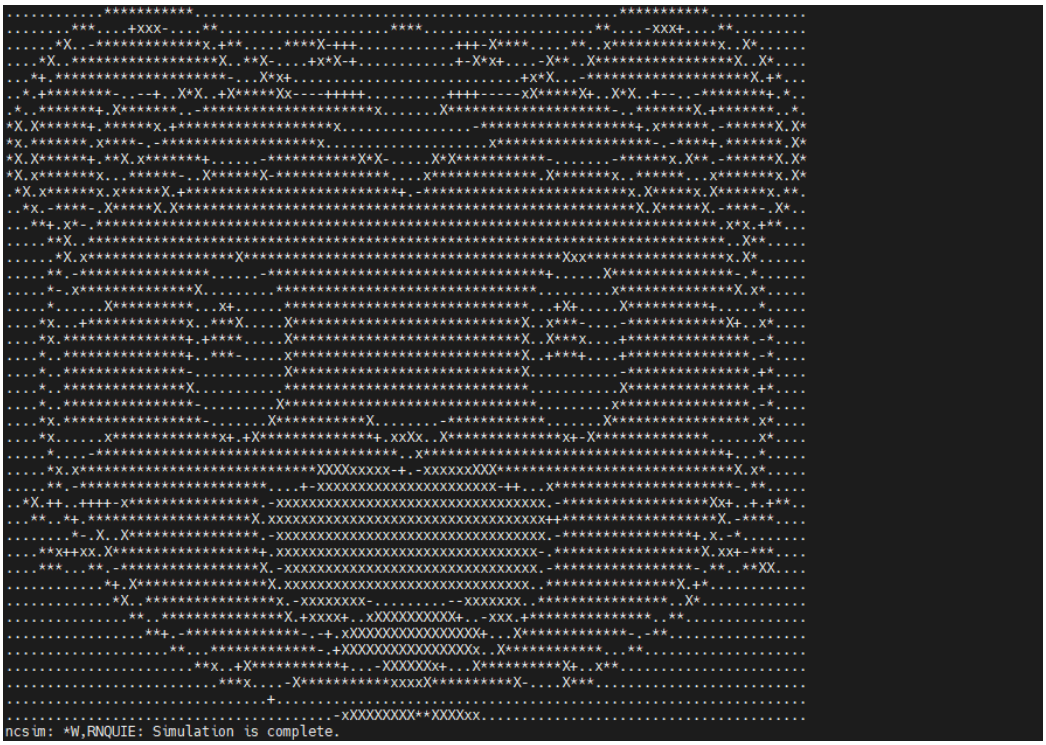
$$P_{sw} = \frac{C_L V_{DD}^2}{T} = f C_L V_{DD}^2$$

e. **Q5: Check whether the synthesis result is correct (screenshot of the result.)**

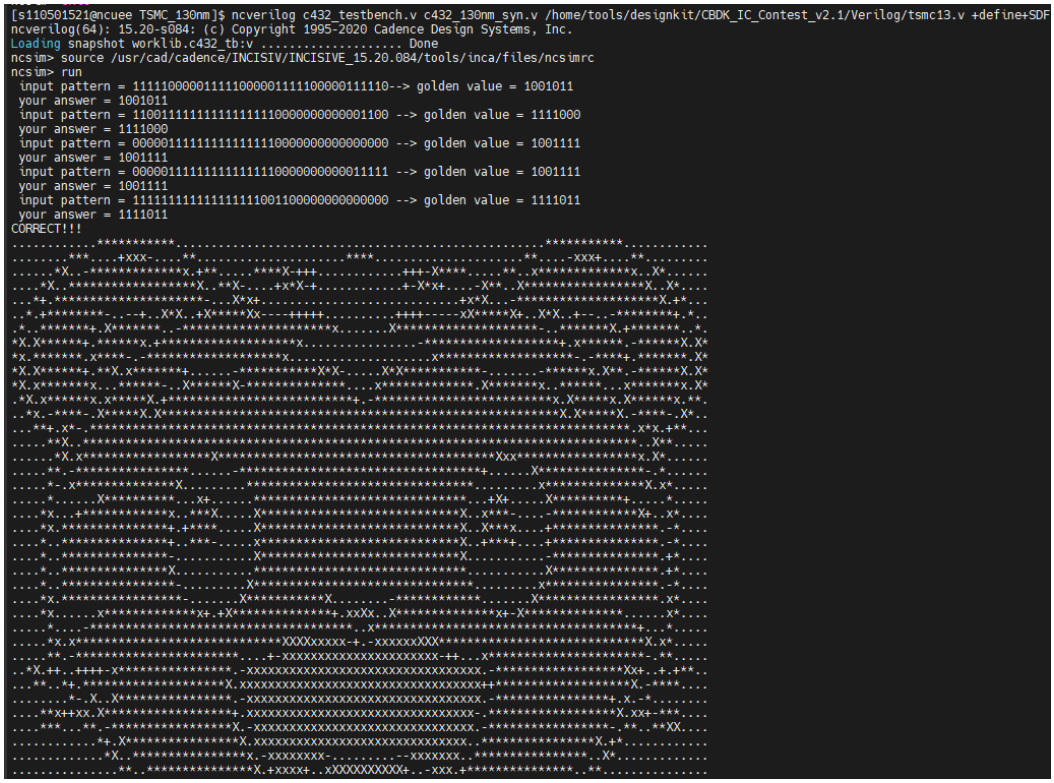
i. **16nm:以合成結果再次做模擬，結果依然正確**

```
ncverilog: *E,FILEMIS: Cannot find the provided file c432_16nm_syn.v-v.
[s110501521@ncuee TSMC_16nm_ADFP]$ ncverilog +access+r c432_testbench.v c432_16nm_syn.v -v /home/ad
fp_tools/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/VERILOG/N16ADFP_StdCell.v
ncverilog(64): 15.20-s084: (c) Copyright 1995-2020 Cadence Design Systems, Inc.
file: c432_testbench.v
        module worklib.c432_tb:v
                errors: 0, warnings: 0
file: c432_16nm_syn.v
        module worklib.c432:v
                errors: 0, warnings: 0
file: /home/adfp_tools/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/VERILOG/N16ADFP_St
dCell.v
        module VERILOG.AO21D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.AOI211D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.AOI21D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.AOI22D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.AOI32D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.IND2D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.IND4D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.INR2D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.INR4D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.INVD1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.IOA21D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.ND2D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.NR2D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.NR3D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.NR4D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.OAI211D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.OAI21D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.OAI22D1BWP20P90:v
                errors: 0, warnings: 0
        module VERILOG.OR2D1BWP20P90:v
                errors: 0, warnings: 0
                Caching library 'VERILOG' ....... Done
                Caching library 'worklib' ....... Done
        Elaborating the design hierarchy:
        Building instance overlay tables: ..................... Done
        Generating native compiled code:
                VERILOG.AO21D1BWP20P90:v <0x3091a48c>
```

```
        Elaborating the design hierarchy:
        Building instance overlay tables: ..................... Done
        Generating native compiled code:
                VERILOG.AO21D1BWP20P90:v <0x3091a48c>
                        streams:   0, words:     0
                VERILOG.AOI211D1BWP20P90:v <0x7207bbfb>
                        streams:   0, words:     0
                VERILOG.AOI21D1BWP20P90:v <0x671424f8>
                        streams:   0, words:     0
                VERILOG.AOI22D1BWP20P90:v <0x055bff5f>
                        streams:   0, words:     0
                VERILOG.AOI32D1BWP20P90:v <0x132f40f1>
                        streams:   0, words:     0
                VERILOG.IOA21D1BWP20P90:v <0x3f6f5ba6>
                        streams:   0, words:     0
                VERILOG.OAI211D1BWP20P90:v <0x026e4cb8>
                        streams:   0, words:     0
                VERILOG.OAI21D1BWP20P90:v <0x320defbe>
                        streams:   0, words:     0
                VERILOG.OAI22D1BWP20P90:v <0x19da8230>
                        streams:   0, words:     0
                worklib.c432_tb:v <0x4b9884ca>
                        streams:  42, words: 107254
        Building instance specific data structures.
        Loading native compiled code:     ..................... Done
        Design hierarchy summary:
                                  Instances  Unique
                Modules:                 83      21
                Primitives:             215       3
                Timing outputs:          81      19
                Registers:               37      37
                Scalar wires:           117       -
                Initial blocks:           1       1
                Pseudo assignments:      36      36
                Simulation timescale:  1ps
        Writing initial simulation snapshot: worklib.c432_tb:v
Loading snapshot worklib.c432_tb:v ..................... Done
ncsim> source /usr/cad/cadence/INCISIV/INCISIVE_15.20.084/tools/inca/files/ncsimrc
ncsim> run
 input pattern = 11111000001111100000111110000011111 0--> golden value = 1001011
 your answer = 1001011
 input pattern = 11001111111111111111100000000000001100 --> golden value = 1111000
 your answer = 1111000
 input pattern = 00000111111111111111100000000000000000 --> golden value = 1001111
 your answer = 1001111
 input pattern = 00000111111111111111100000000000011111 --> golden value = 1001111
 your answer = 1001111
 input pattern = 11111111111111111100110000000000000000 --> golden value = 1111011
 your answer = 1111011
CORRECT!!!
```



ii. **130nm: 以合成結果再次做模擬，結果依然正確**



2. The difficulties I encountered and how I solved the problems

這次 bonus 是在 228 連假做的，不過沒想到在做 130nm 合成的時候碰到以合成結果驗證有出錯的問題，一開始以為是我的 c432.v 的 coding style 有問題或是有錯誤，不過和同學核對後發現應該是沒問題，於是就詢問祝教官於 design compiler 的設定方式。**助教也非常**

迅速就解決了，真的是辛苦卓助教了，這次操作下來，我覺得 design compiler 在使用上其實不像 vivado 這麼直觀(雖然 vivsdo 我也是用了很久才比較熟一點)，而且 design compiler 有時候會碰到在 behavioral 的模擬沒有問題，但由 design compiler 做完邏輯化簡，合成出後再以相同的 testbench 測試會有問題的情況，**原因應該是 design compiler 本身對 coding style 的要求更高，對於 RTL code 在描述電路行為上更不容許有任何模糊之處(相比 vivado，畢竟 vivado 的合成是為了 FPGA，FPGA 的運作本質上是用 memory 實現 look up table 來 mapping 出我們 RTL 描述的電路邏輯，所以還是比較寬鬆一些，因為不是真的會合成出電路)，整體而言 design compiler 的操作以及如何符合 design compiler 的要求我覺得是比較花心力的地方**。

3. What I've learn from this assignment

a. 使用 design compiler 的方式: **助教教得好仔細，真的很感謝!**

b. RTL code 在 coding style 上要注意的細節避免 design compiler 合成出非預期結果

c. 突然覺得 vivado 蠻好用的，不用反覆在 testbench 上加上 display 的語法，之後修課寫作業也會盡量搭配 vivado 做模擬驗證 functionality。

4. Suggestions of this bonus assignment

我覺得這個 bonus project 是蠻好的練習，不過我個人會很想知道關於 design compiler 的合成機制，畢竟我常常碰到在 behavioral simulation 結果正確的情況下，synthesis 後的.v file 的模擬結果出錯，這邊應該是 coding style 的問題(但已經沒有 multi-driven 或是其他明顯 coding style 的問題了)，總覺得要稍微弄懂 design compiler 的合成機制，不然之後寫作業還是會遇到。這次真的很感謝卓助教的幫助，而且卓助教居然在早上四點回我 email，真的是辛苦助教了!祝助教後續的研究與求職都順利!(這門課真的辛苦你們了 QQ)。